

Week 5 – Practice Session

October 5–9, 2026

Function Transformations

In this section, we look at some basic ways to change the graph of a function. These transformations help us sketch the graph of a new function using the graph of a function we already know.

- **Vertical shift:** $f(x) + k$ moves the graph k units upward, while $f(x) - k$ moves it k units downward, where $k > 0$.
- **Horizontal shift:** $f(x - h)$ moves the graph h units to the right, while $f(x + h)$ moves it h units to the left, where $h > 0$.
- **Vertical stretch or compression:** $af(x)$ stretches the graph vertically if $|a| > 1$ and compresses it vertically if $0 < |a| < 1$. If $a < 0$, the graph is also reflected across the x -axis.
- **Horizontal stretch or compression:** $f(bx)$ compresses the graph horizontally if $|b| > 1$ and stretches it horizontally if $0 < |b| < 1$. If $b < 0$, the graph is also reflected across the y -axis.

A function can be changed by applying several transformations one after another. It is often helpful to do this step by step and sketch the graph after each transformation. If we draw several graphs in the same coordinate system, using different colors can make them easier to distinguish.

Quadratic Function Transformations

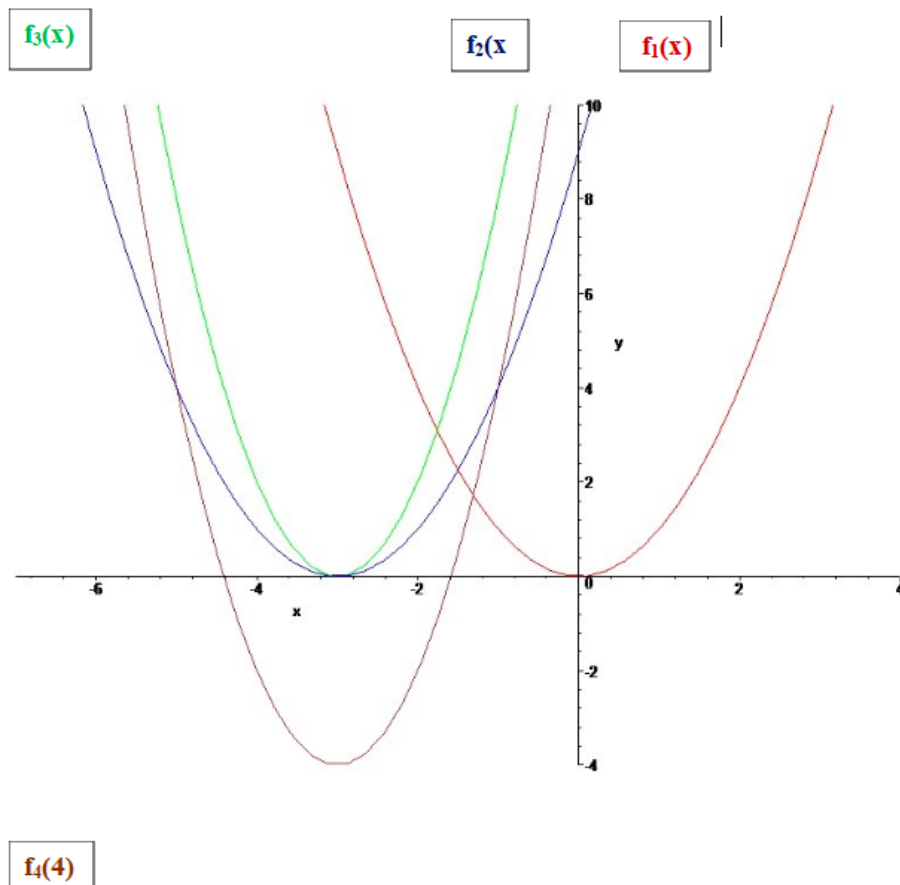
Example 1. Sketch the graph of the following function and give its largest possible domain and range in interval notation:

$$f(x) = 2(x + 3)^2 - 4.$$

Solution. Starting from the basic function $f_1(x) = x^2$, we obtain

$$f_1(x) = x^2 \longrightarrow f_2(x) = (x+3)^2 \longrightarrow f_3(x) = 2(x+3)^2 \longrightarrow f_4(x) = f(x) = 2(x+3)^2 - 4.$$

The following figure shows the graphs of $f_1(x)$, $f_2(x)$, $f_3(x)$, and $f(x) = f_4(x)$.



Properties of $f(x)$:

Extreme value: minimum at $x = -3$, with $y = -4$.

Domain: $D_f = (-\infty, +\infty)$.

Range: $R_f = [-4, +\infty)$.

Roots: setting $f(x) = 0$ gives $2(x + 3)^2 - 4 = 0$, so $x_1 = -3 - \sqrt{2}$, $x_2 = -3 + \sqrt{2}$.

y -intercept: setting $x = 0$ gives $y = 14$.

Monotonicity: the function is strictly decreasing on $(-\infty, -3]$ and strictly increasing on $[-3, +\infty)$.

Quadratic Functions

For each function, sketch its graph and give its largest possible domain and range in interval notation.

1. $y = 0.5(x - 5)^2$
2. $l(x) = -x^2 - 6x - 8$
3. $f(x) = -(x + 2)(x - 3)$

Absolute Value Functions

Consider the basic function

$$f(x) = |x|.$$

Sketch the graph of each of the following functions and give its largest possible domain and range in interval notation.

4. $f(x) = -4|x + 6|$
5. $y = 2|x - 3| - 4$
6. $g(x) = |2x + 4| + 3$

Exponential Functions

For each function, sketch its graph and give its largest possible domain and range in interval notation.

7. $h(x) = 3^{x+2}$
8. $z(x) = 3^{x+2} - 1$
9. $q(x) = 2^{x-3} + 2$

Logarithmic Functions

For each function, sketch its graph and give its largest possible domain and range in interval notation.

9. $k(x) = \log_2(x + 3)$
10. $u(x) = \log_2(x + 3) + 2$

Rational Functions

Consider the basic function

$$p(x) = \frac{1}{x}.$$

Sketch the graph of each of the following functions and give its largest possible domain and range in interval notation.

11. $r(x) = \frac{1}{x-3}$
12. $y = \frac{1}{x-3} + 2$

Square Root Functions

Consider the basic function

$$m(x) = \sqrt{x}.$$

Sketch the graph of each of the following functions and give its largest possible domain and range in interval notation.

13. $y = \sqrt{x - 3}$

14. $j(x) = \sqrt{x - 3} - 5$